

AKASH SHEKHAR PATIL

AI Engineer | Generative AI, RAG, LLM Applications, Python Backend, Cloud

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PROFESSIONAL SUMMARY

AI Engineer and Python Backend Developer with 5+ years of experience building production-grade generative AI applications, RESTful APIs, and cloud-native automation. Builds LLM applications and APIs with Python, FastAPI, Flask, and LangChain. Designed and deployed end-to-end retrieval-augmented generation (RAG) systems covering document ingestion, structure-aware chunking, embeddings, hybrid retrieval, reranking, prompt orchestration, source citations, guardrails, and evaluation using Azure OpenAI, LangChain, Hugging Face, Pinecone, and Azure AI Search. Proven track record delivering scalable serverless backends on AWS, migrating and optimising enterprise databases, and leading client-facing engagements for global customers. Microsoft Certified: Azure Developer Associate (AZ-204).

TECHNICAL SKILLS

Generative AI & LLM: Retrieval-Augmented Generation (RAG), LangChain, AI agents, large language models (LLM), Hugging Face models and transformers, prompt engineering, prompt orchestration, context assembly, conversation state management, grounded generation, source citations, hallucination mitigation, LLM guardrails, prompt-injection defence, out-of-scope refusal, low-confidence fallback

Retrieval & Vector Search: Pinecone, Azure AI Search, vector databases, embeddings, Hugging Face sentence transformers, semantic search, hybrid retrieval, BM25 keyword search, dense retrieval, top-K search, metadata filtering, structure-aware chunking, cross-encoder and LLM reranking

AI Evaluation: Recall@K, Precision@K, MRR, answer groundedness, citation coverage, hallucination rate, response latency, user-feedback loops, audit logging

Languages: Python, SQL, R

Frameworks & Libraries: FastAPI, Flask, Django, LangChain, Hugging Face Transformers, REST API design, Pandas, NumPy, unit testing, regression testing

Microsoft Azure: Azure OpenAI, Azure AI Search, Azure Functions, Blob Storage, Queues, API Management, Power Apps, Power Automate

Amazon Web Services: AWS Lambda, API Gateway, EC2, S3, RDS, ElastiCache, Redis, EventBridge, KMS, CloudWatch

Databases: Microsoft SQL Server, MySQL, Oracle BRM, relational data modelling, query migration and optimisation

DevOps & Tools: Docker, Kubernetes, Git, GitLab, Excel, Agile and Scrum delivery

PROFESSIONAL EXPERIENCE

Senior Python Developer / AI Engineer | LTIMindtree

Sep 2024 - Present

Client: Microsoft (Lumenisity Engagement) | Pune, India

- Designed and implemented an AI agent for enterprise Standard Operating Procedures (SOPs) that lets employees retrieve step-by-step, source-grounded guidance from internal documents through natural-language questions.
- Built an end-to-end RAG ingestion pipeline in Python to extract and clean PDF and DOCX content, apply structure-aware chunking, enrich chunks with document and section metadata, generate embeddings with Hugging Face and Azure OpenAI models, and index content into Pinecone and Azure AI Search.
- Implemented hybrid retrieval combining BM25 keyword search with dense vector similarity, layered with metadata filtering and reranking to improve relevance for procedure names, equipment terminology, and error codes.
- Developed prompt orchestration using LangChain that assembled ranked context, maintained conversation state, generated concise procedural answers, and returned document, section, and page-level citations for full traceability.
- Engineered enterprise guardrails including prompt-injection checks, out-of-scope refusal, restricted-topic filtering, source-grounding enforcement, low-confidence fallback, input and output validation, and audit logging.
- Established a RAG evaluation framework measuring Recall@K, Precision@K, MRR, answer groundedness, citation coverage, hallucination rate, and response latency, using the results to tune retrieval and prompt strategy.
- Exposed the assistant through secure Python FastAPI and Flask services, integrating Azure Blob Storage, Queues, and Azure Functions for document storage, asynchronous processing, telemetry, and deployment.
- Built Flask APIs for a fibre inspection application that processed uploaded images, generated 2D fibre visualisations, and detected ring defects to support automated quality assessment.

- Led migration of Python data-access modules from MySQL to Microsoft SQL Server, refactoring queries and completing unit and regression testing to preserve accuracy and application stability.
- Designed and maintained 10+ Power Apps serving 83+ active users and automated Excel-based workflows with Power Automate, reducing manual effort by an estimated 60-70%.
- Acted as primary client-facing developer for the Microsoft engagement, leading requirement workshops, demos, feedback cycles, release coordination, and stakeholder sign-off.

Environment: *Python, FastAPI, Flask, LangChain, Azure OpenAI, Hugging Face, Pinecone, Azure AI Search, RAG, BM25, embeddings, vector databases, reranking, Azure Functions, Blob Storage, Queues, SQL Server, Power Apps, Power Automate, Docker, Kubernetes, Git*

Python Developer | Edwisely

Mar 2022 - Nov 2023

Assessment Platform, Serverless Migration & Database Re-architecture | Remote, India

- Designed and developed REST APIs using Python, Flask, AWS Lambda, and API Gateway for a multiple-choice assessment platform consumed by web and mobile clients.
- Migrated Flask-based endpoints to Python AWS Lambda functions exposed through API Gateway, delivering a scalable serverless architecture that reduced backend API response times by 50%.
- Improved MCQ test-interface performance by optimising the database layer on AWS RDS and introducing Redis and ElastiCache caching, significantly increasing responsiveness and scalability of the module.
- Led re-architecture of the SQL database on AWS RDS and modified dependent APIs to support the new schema, improving scalability and maintainability across the platform.
- Translated functional requirements into backend business logic, reusable services, database interactions, and testable API contracts; managed all source changes through GitLab.
- Mentored and coordinated a cross-functional team of four engineers, partnering with frontend developers to integrate APIs and resolve end-to-end delivery issues.
- Monitored serverless workloads and application behaviour with Amazon CloudWatch, investigated production failures, and supported stable releases within Agile ceremonies.

Environment: *Python, Flask, SQL, AWS Lambda, API Gateway, RDS, EC2, ElastiCache, Redis, EventBridge, CloudWatch, GitLab*

Associate Software Engineer | Accenture

Aug 2019 - Dec 2020

Telecom Billing - Oracle BRM Data Engineering & Automation | Pune, India

- Built a Python and SQL based data monitoring tool that identified daily data-synchronisation failures across Oracle BRM billing tables and automatically inserted missing records, preventing incorrect customer invoicing and significant revenue leakage for the client.
- Automated daily and monthly aggregation of customer internet usage data to feed downstream billing operations, replacing manual reconciliation effort.
- Optimised the Oracle BRM file-processing pipeline by developing a Python pre-processing utility that restructured the flow of files from upstream systems, reducing total process runtime by 10%.
- Replaced a legacy extraction system with a Python application for database extraction and flat-file encoding, improving maintainability and processing reliability.
- Collaborated with client stakeholders in an Agile delivery model on requirement analysis, testing, and production support for a large-scale telecom billing platform.

Environment: *Python, SQL, Oracle BRM, Excel*

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate (AZ-204)
- GitHub Copilot Fundamentals
- Python Basics - KodeKloud

EDUCATION

Bachelor of Engineering (B.E.), Computer Engineering

Jul 2015 - May 2019

Smt. Kashibai Navale College of Engineering, Savitribai Phule Pune University, Pune, India - Distinction, 7.98 GPA